🎓 Cambridge AI Academy

INTENSIVE MASTERCLASS · 120 MINUTES

Building AI Agents for Real Business Growth

🧠 Architecture | 📚 Memory | ⚒️ Multi-agent teams | 🚀 Live deployment

WELCOME · 00:00

Before we start



Bring a computer

You will build on your own laptop today.



Connect to the Wi-Fi

Details on the next slide.



Open the Canva Whiteboard

Use the session links we share in the room.



Break at 55 min

5 minutes, then Module 3.

Get connected

NETWORK	[Network name]
PASSWORD	[Password]

Passwords are case-sensitive. Ask if you have trouble getting online.

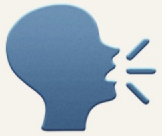
THE PLAN

120-minute master plan

00:00–00:10	👋	**Welcome & setup**	Wi-Fi, whiteboard, how we learn
00:10–00:35	🧠	**Module 1 · Agent architecture**	Prompts → execution loops, ReAct
00:35–00:55	📚	**Module 2 · Memory & grounding**	RAG, citations, memory types
00:55–01:00	☕	**Break**	5 minutes
01:00–01:30	🛠️	**Module 3 · Multi-agent teams**	n8n, MCP, supervisor-worker
01:30–01:50	🚀	**Module 4 · Deploy & monetise**	Fly.io, Key Vault, Stripe
01:50–02:00	✅	**Launch & wrap-up**	Checklist, go live, next steps

WELCOME · HOW WE LEARN

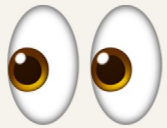
Tell · Show · Do · Apply



01

Tell

I explain the idea and why it matters.



02

Show

I demo it live so you see it working.



03

Do

You try it yourself, with help on hand.



04

Apply

You use it on your own problem or project.

HOW TODAY WORKS

We run this session on a Canva Whiteboard

🔗 Join the board

Open the link: [board link]. No account needed to view.

🟨 Add a sticky note

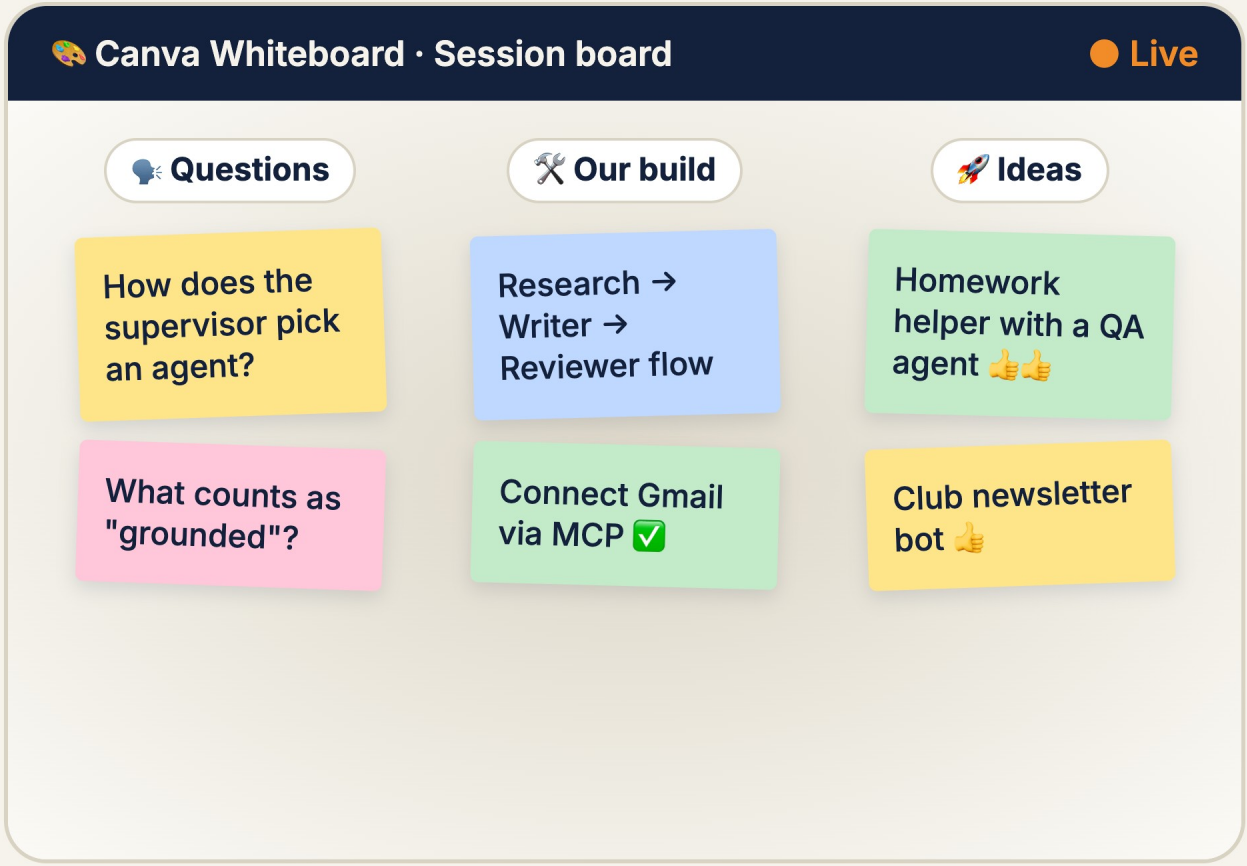
Put your name on one so we know you're in.

🧩 Work in your lane

Each activity has its own section on the board.

👍 Vote and react

Add emoji to the ideas you like best.



WHY NOW · FROM CI TO CD

AI integrates. People pilot delivery.

THEN · CI



Manual integration

People spend time merging changes, resolving conflicts and moving work through the pipeline.

NOW · CD



AI-enabled delivery

AI handles integration: assembling changes, testing them and surfacing what needs attention.

👨‍✈️ People become pilots: guiding the system, supervising outcomes, making the calls and driving delivery 🚀

THE BIG PICTURE · TELL 🗣️

8 layers of an AI business bot

LAYER 1

🧠 Model layer

Claude to reason and use tools; Groq for fast inference.

LAYER 2

🧩 AI harness

MCP, LangGraph, CrewAI, Agents SDK or custom code.

LAYER 3

🛠️ Business tools

CRM, email, calendar, payments, behind approval gates.

LAYER 4

📚 Knowledge

Search, embeddings, vector DB, reranking, citations.

LAYER 5

🚀 DevOps

GitHub, tests, Docker, staging, canary, rollback.

LAYER 6

🔒 Security

Least privilege, SSO, secrets, sandboxing, audit trails.

LAYER 7

🧪 Examples

GroqBot, Muse, Dots and custom Claude + MCP builds.

LAYER 8

👨‍✈️ People as pilots

Set objectives, supervise, review risk, own outcomes.

00:10 – 00:35

MODULE 1

Visual Agent Architecture & Tool Creation

Goal: move from single-turn chat prompts to deterministic execution loops.

4-Step
Perception, reasoning, action, feedback

100%
Deterministic tool JSON schemas

1-on-1
Laptop environment check by Marianna

MODULE 1 · TELL 🗣️

From one-off prompts to execution loops

OLD WAY

💬 Single-turn chat prompts

Prompt → AI response → end

- ✖ One-off inputs and outputs
- ✖ Inconsistent, unpredictable results
- ✖ Relies only on next-token prediction
- ✖ No feedback or self-correction

NEW WAY

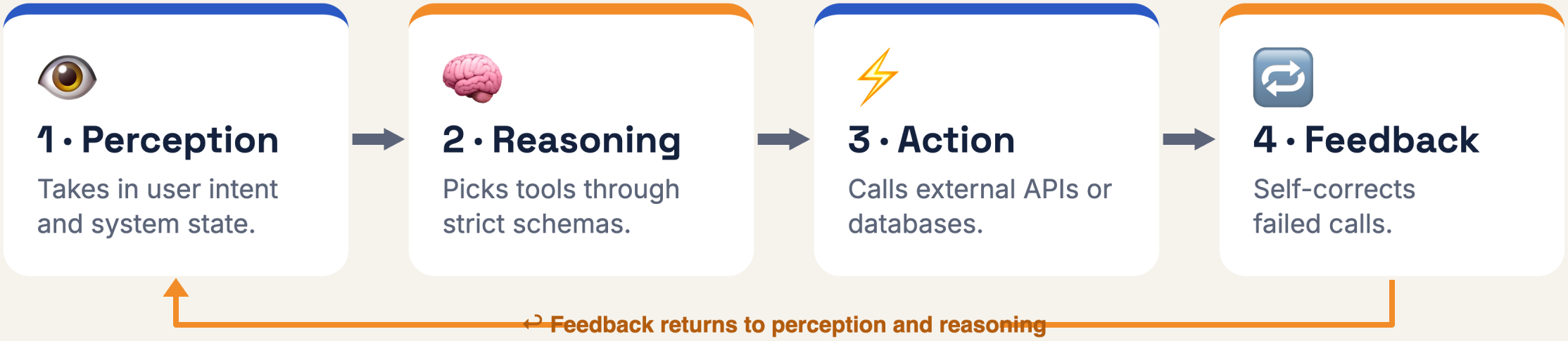
🔁 Deterministic execution loops

Goal → act → validate → refine ↻

- ✔ Continuous execution loops
- ✔ Consistent, repeatable, predictable
- ✔ Systemic error handling and validation
- ✔ Goal-oriented and self-correcting

MODULE 1 · SHOW 👀

The ReAct decision loop



In practice: 🎯 Goal → 🧠 Reason → 🛠️ Act → 👀 Observe → ✅ Verify → 🔁 Repeat until done or a human is needed

MODULE 1 · APPLY 🚀 · 5 MINUTES

Give 10 ideas for your bots

Add one sticky note per idea in the 🚀 Ideas lane of the whiteboard.

🔁

What do you repeat?

A task you do every week.

🛠️

Which tools?

Email, calendar, sheets, CRM...

👨‍✈️

Who checks it?

Where a human approves.

00:35 – 00:55

MODULE 2

Business Document Memory & Grounding

Ground every answer in trusted business documents.



Fewer hallucinations



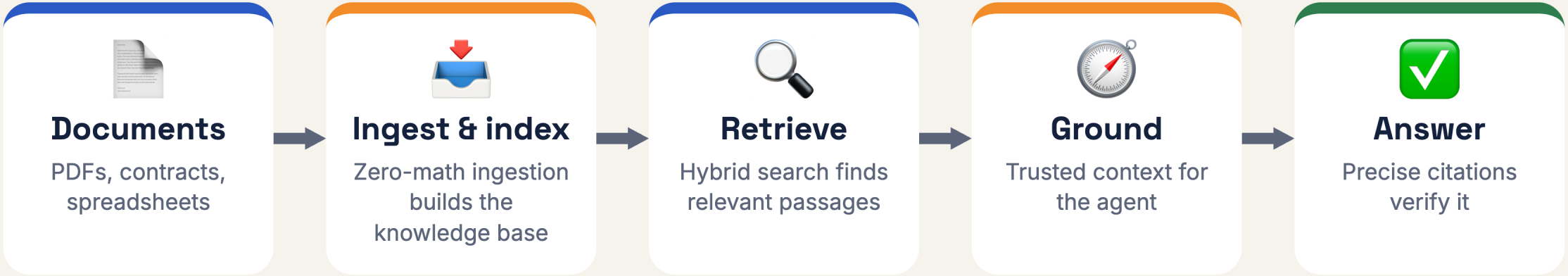
More accurate answers



Answers you can audit

MODULE 2 · TELL 🗣️

How RAG grounds your agent



🖱️ Zero-math ingestion

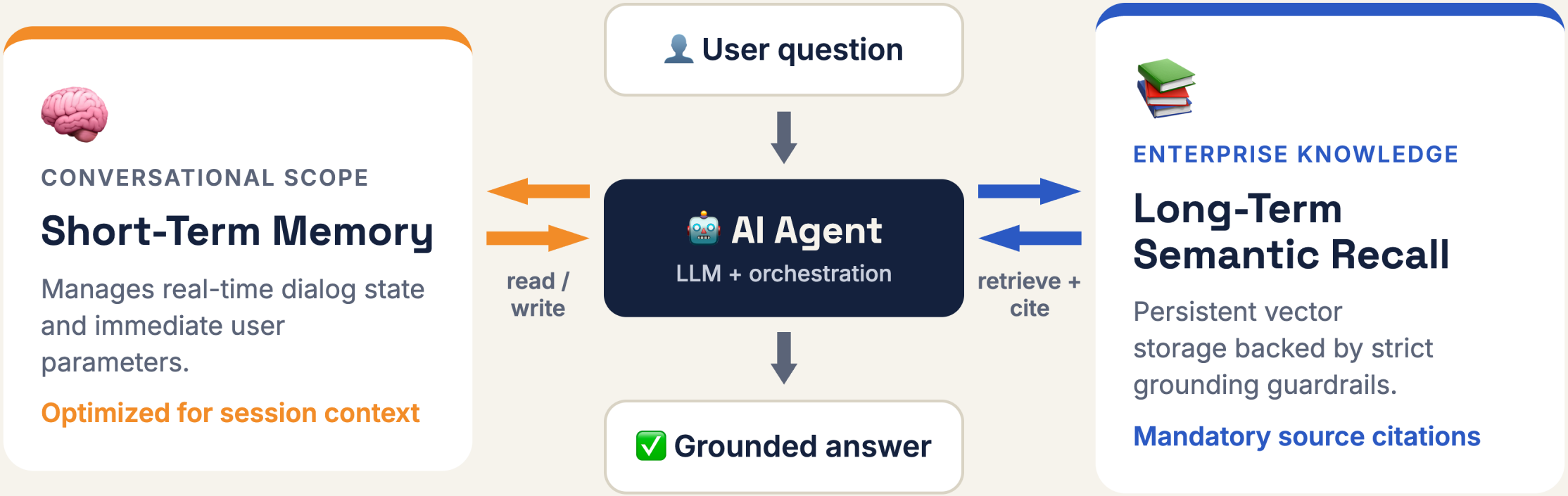
Drag and drop PDFs, contracts and spreadsheets.

🔎 Hybrid search & citations

Keyword plus semantic ranking, with sources checked.

MODULE 2 · SHOW 👀

Enterprise Memory Architectures





5-minute break

Back at 01:00 for Module 3: multi-agent teams

01:00 – 01:30

MODULE 3

Multi-Agent Teams with n8n & MCP

Split work across specialised agents that share the same tools.

🧭 Routing

A supervisor delegates complex requests to specialised agents

🔌 MCP

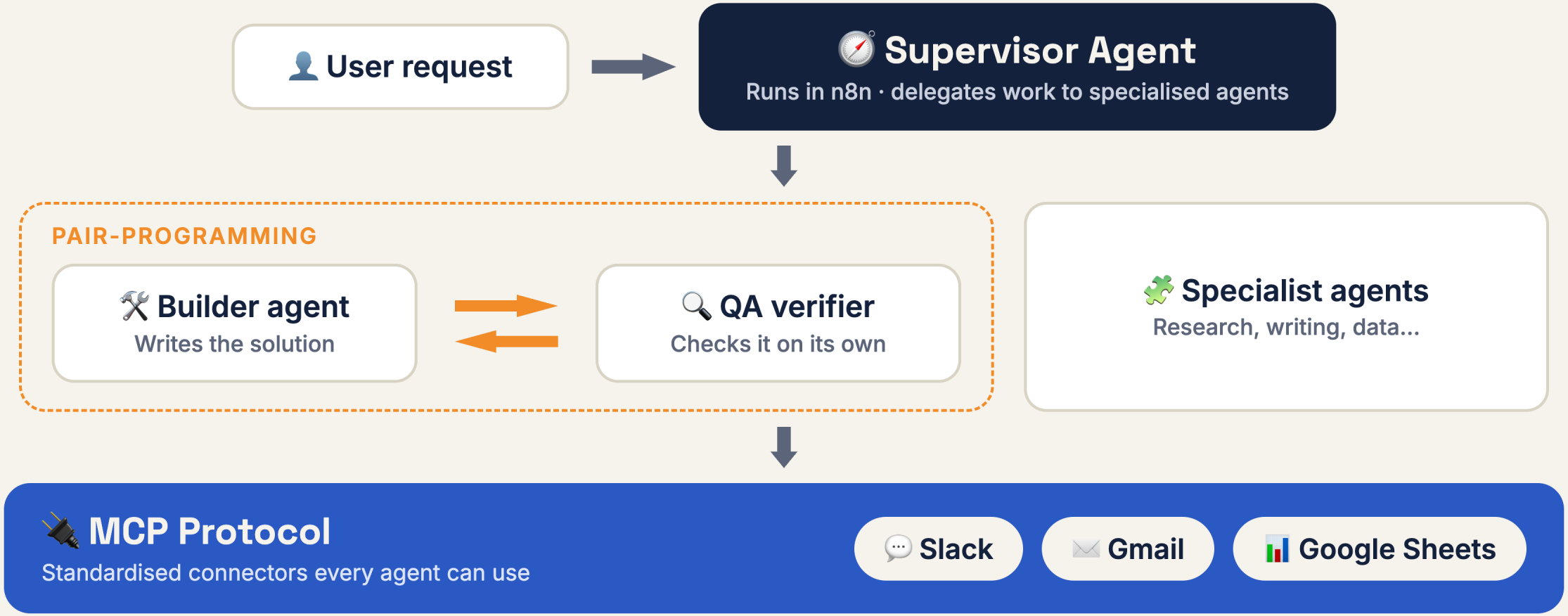
Standard connectors for Slack, Gmail and Google Sheets

👥 Pairing

Builder agents paired with autonomous QA verifiers

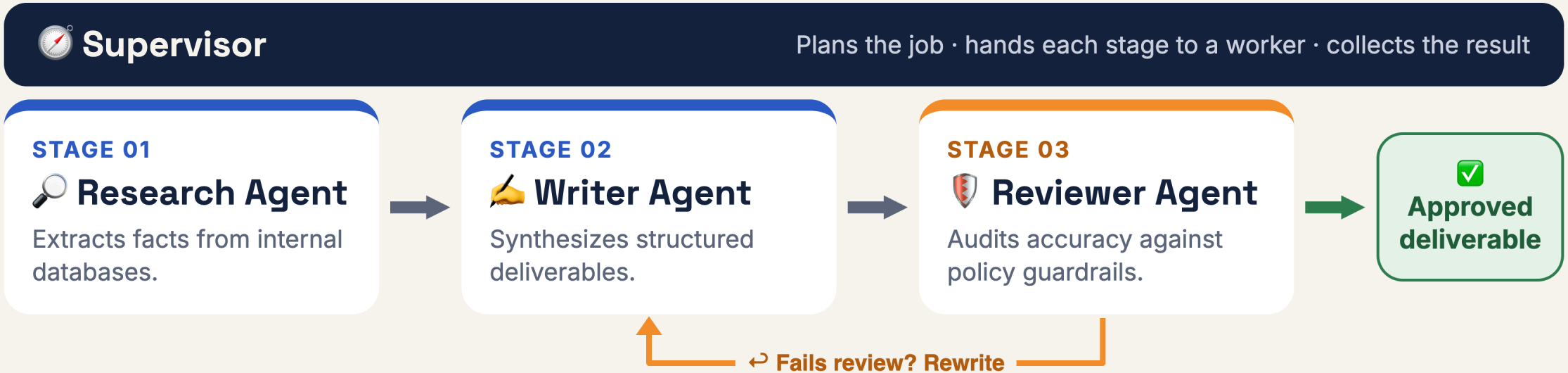
MODULE 3 · SHOW 👀

Multi-Agent Teams with n8n & MCP



MODULE 3 · TELL 🗣️

Supervisor-Worker Execution Flow



Why split it?

🎯 **One job each.** Focused agents give better output.

📌 **Facts first.** Writing is built on real data.

🚦 **Checked before it ships.** Nothing skips review.

MODULE 3 · SHOW 👀 THEN DO 🛠️

Live demo, then build your team

WATCH · 10 MIN

👀 Real bots at work

📋 Our bot catalogue

🔀 An n8n supervisor flow

💻 An agent fixing code in the terminal

BUILD · 15 MIN

🛠️ Your turn

1. Pick your top idea from the board
2. Draw a supervisor and 2–3 workers
3. Connect one MCP tool: Slack, Gmail or Sheets

01:30 – 01:50

MODULE 4

Cloud Deployment & Live Monetization

1-Click

Edge deployment to London servers with live Stripe checkout

Leave the session with a public production URL.

MODULE 4 · TELL 🗣️

Where to deploy

	GitHub Pages	Fly.io ★	Railway
BEST FOR	Static websites (HTML, CSS, JS)	Full-stack apps and APIs	Full-stack apps and databases
PRICING	Free (1 GB repo, 100 GB/month bandwidth)	Pay-as-you-go, free trial	Pay-as-you-go, Hobby plan
DATABASE	None, frontend only	Postgres, Redis, SQLite, edge DBs	Postgres, Redis, MongoDB, MySQL
HIGHLIGHTS	Git-based deploys, custom domains, HTTPS	Docker deploys worldwide, scales to zero	1-click templates, auto SSL and secrets

★ We deploy to Fly.io today: it runs full agents and APIs close to London users.

MODULE 4 · SHOW 👀

Edge runtime & revenue stack



Fly.io London Edge

Low-latency runs on high-availability containers



Azure Key Vault

Isolated secrets for enterprise API credentials



Stripe metered payments

Pay-per-execution billing for agent calls

↺ Fast and secure revenue loop

LAUNCH · DO 🛠️ · 01:50

Live launch checklist

☐ 01 · INPUT GATE



Error handling

Malformed inputs handled predictably

☐ 02 · REASONING



Citation guardrails

Answers must cite documents

☐ 03 · BILLING



Stripe webhooks

Payment events verified

☐ 04 · LIVE RUNTIME



Telemetry & caps

Trajectory logs and token limits on

All four ticked? Launch status: go for flight 🚀

LOOKING AHEAD

AI creating AI

🔨

Era 1 · Human-made tool

For millennia people designed and built every tool by hand.



The bridge

Tools became digital and adaptive: they learn and optimise.

🤖

Era 2 · Tools build tools

AI systems now analyse problems and create new tools themselves.

We are no longer only building objects. We build the intelligence that builds the next tools.

SESSION COMPLETE

Your agent is live 🎉

Test your live endpoint, verify your first payment and keep building.

🗣️ Tell → 👀 Show → 🛠️ Do → 🚀 Apply → 🌐 Live

👥 Marianna & team

🌐 Live edge endpoint

APPENDIX

Go deeper after class

These detailed visuals from the original deck are for self-study.

🎬 How LLMs read tokens

Video explainer: words become numbers, then meaning.

🗺️ How RAG works, in full

The 11-step infographic from ingestion to answer.

🌀 Building tools that build tools

A Lacanian framework: the real, the imaginary, the symbolic.

🔁 The agentic workflow loop

Goal, reason, act, observe, verify, repeat.